



Quantum computing

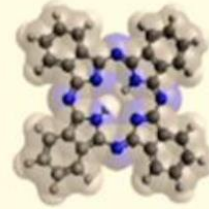
Taras Holotyak

Applications of quantum computers

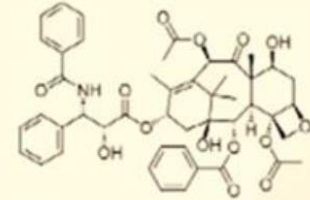
TIME
"Quantum Will
Change
Everything"



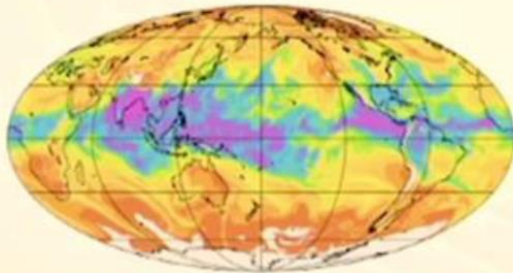
Travel & Logistics



Chemistry



Pharmacology



Climate Modeling



Financial Analysis



Cryptography

Applications of quantum computers

Application 1. Quantum Factoring

P. Shor (1994)

A quantum computer can factor numbers
exponentially faster than classical computers

$15 = 3 \times 5$ (...easy)

$38647884621009387621432325631 = ? \times ?$

Importance: cryptanalysis

public key cryptography relies on
inability to factor large numbers

Best classical algorithm: 10^{24} steps	Shor's quantum algorithm: 10^{10} steps
On classical THz computer: 150,000 years	On quantum THz computer: <1 second

Applications of quantum computers

Application 2: Quantum Search

L. Grover (1997)

A quantum computer can find a marked entry in an unsorted database **quadratically faster** than classical computers

(e.g., given a phone number, finding the owner's name in a phonebook)

Importance: “satisfiability” problems

- fast searching of big data
- inverting complex functions
- determining the median or other global properties of data
- pattern recognition; machine vision

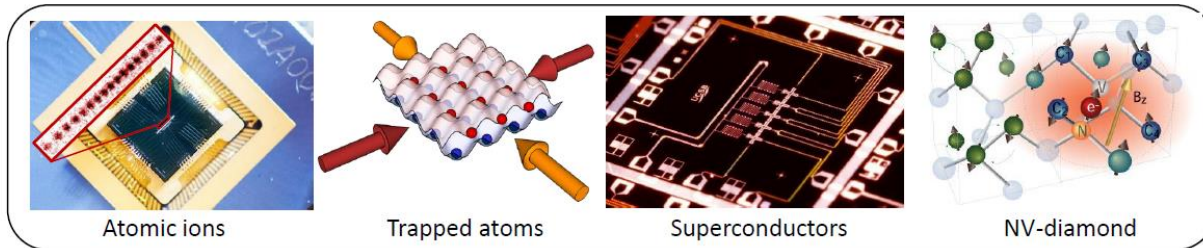
Applications of quantum computers

Application 3: Quantum Simulation

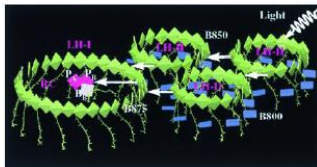
Quantum modelling is hard: N quantum systems require solution to 2^N coupled eqns

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi$$

Alternative approach: Implement model of interacting system on a *quantum simulator*, or “standard” set of qubits with programmable interactions



Quantum Material Design Understand exotic material properties or design new quantum materials from the bottom up



Energy and Light Harvesting Use quantum simulator to program QCD lattice gauge theories, test ideas connecting cosmology to information theory (AdS-CFT etc..)

Quantum Field Theories Program QCD lattice gauge theories, test ideas connecting cosmology to information theory (AdS-CFT etc..)

Applications of quantum computers

Application 4: Quantum Optimization

Minimizing complex (nonlinear) functions by “simultaneously sampling” entire space through quantum superposition

Relevant to

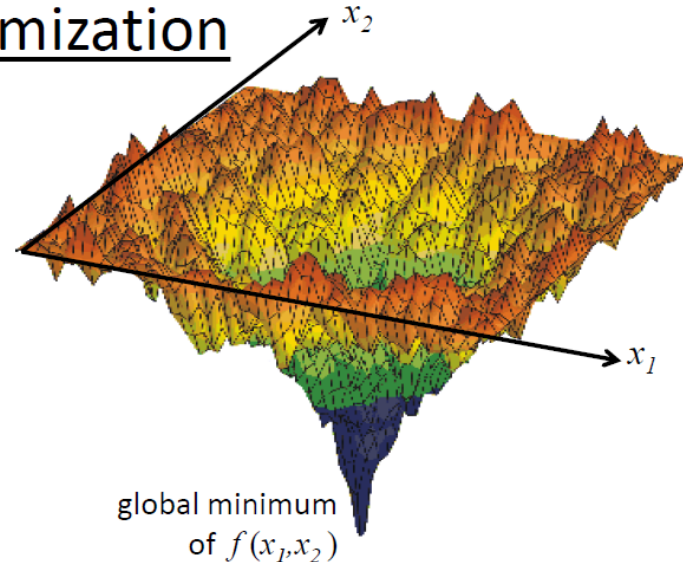
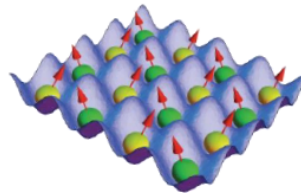
- Logistics
- Operations Research
- VLSI design
- Finance

Example: quadratic optimization

Minimize

$$f(x_1, x_2, \dots) = \sum_{i < j} q_{ij} x_i x_j + \sum_i c_i x_i$$

this function maps to energy of quantum magnetic network

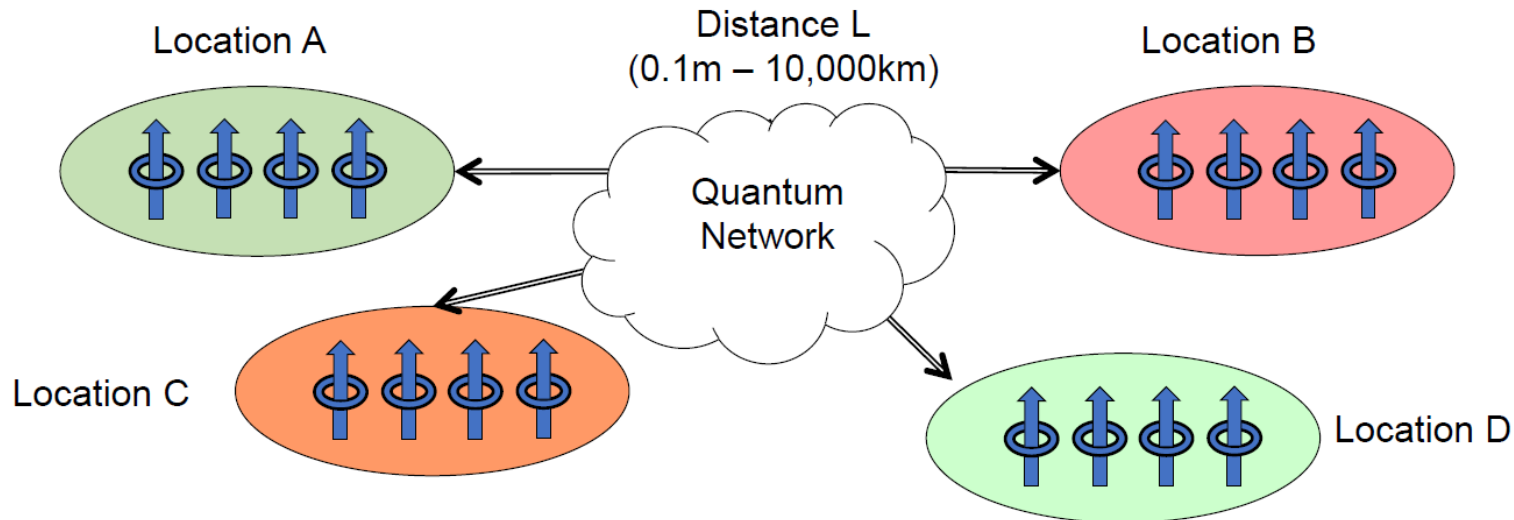


Killer Application?

- could crack a large class of intractable problems: factoring, “traveling salesman” problem, etc..
- BUT not known if there is always a quantum speedup

Applications of quantum computers

Application 5: Quantum Networks



Uses of a quantum network

- Secret key generation: cryptography
- Certifiable random number generation
- Quantum repeaters ("amplifiers")
- Distributed quantum entanglement for optimal decision making
- *Large-scale quantum computing*

Applications of quantum computers

Before getting into the details of quantum computing proper, we will look at some other aspects of quantum information processing, which have remarkable results that cannot be achieved classically, even in principle. Specifically, we will look at:

- Using *entanglement as a resource*, in teleportation and superdense coding.
- Using quantum phenomena to achieve information theoretically (rather than computationally) secure communications.

Applications of quantum computers

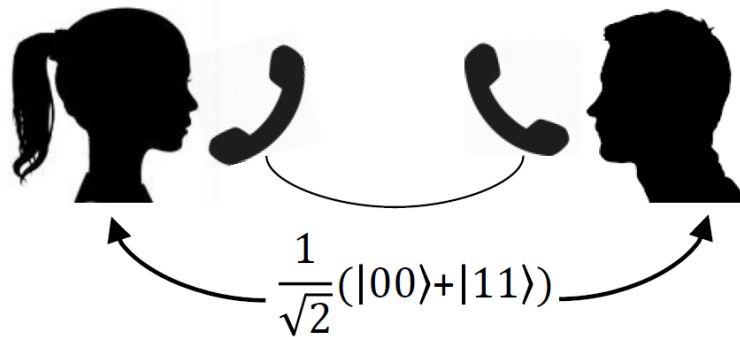
Case Study 1: Quantum communications

- **Teleportation:** using shared entanglement as a resource that allows a qubit to be transmitted using two bits.
- **Superdense coding:** using shared entanglement as a resource that allows two bits to be transmitted using a single qubit.
- **Quantum key distribution:** creating a one-time pad without Alice and Bob meeting or sharing an absolutely secure communication channel.

Applications of quantum computers

Alice and Bob revisited

Alice and Bob once again share an entangled pair $((1/\sqrt{2})(|00\rangle + |11\rangle))$. Previously we saw that they couldn't use this alone for signalling, so we will also give them a communication channel.



We will now see how they can:

1. Use the shared entanglement and **two bits** of classical information to **transfer one qubit** (teleportation).
2. Use the shared entanglement and **one qubit** of quantum information to **transfer two classical bits** (superdense coding).

Applications of quantum computers

Quantum Teleportation

Transporting Qubits

According to the no-cloning theorem, qubits cannot be duplicated.

Is there a way to transmit qubits from one point to another without destroying superposition and phase information?

$$|\Psi\rangle = \alpha |0\rangle + \beta |1\rangle$$



Alice



Bob

Applications of quantum computers

Quantum Teleportation

Quantum Teleportation enables the transfer of qubits, while perfectly preserving state information.

Allows for transmission over great distances...think from Earth to space!

Does not involve:

- Moving matter from one point to another via dematerialization/materialization
- Travelling forward or backward in time

Applications of quantum computers

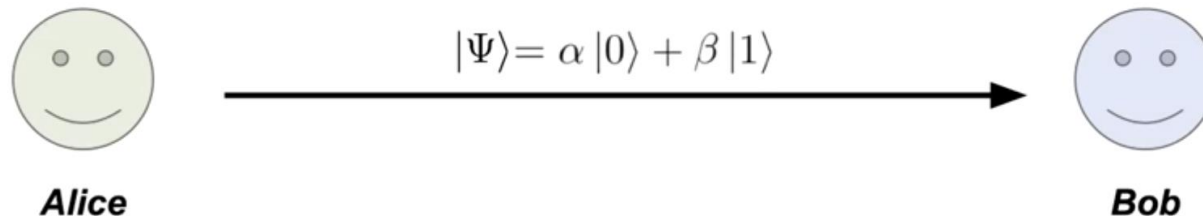
Quantum Teleportation

Requirements for Quantum Teleportation

A qubit can be perfectly transmitted using 3 qubits and 2 classical bits.

Teleportation needs:

- Two qubits that are entangled...each party has one half of the entangled pair
- A 'message' qubit that will be sent from one point to another
- A classical communication line between both parties for the transmission of two classical bits



Note: Since quantum teleportation needs classical communication, it cannot be faster than the speed of light!

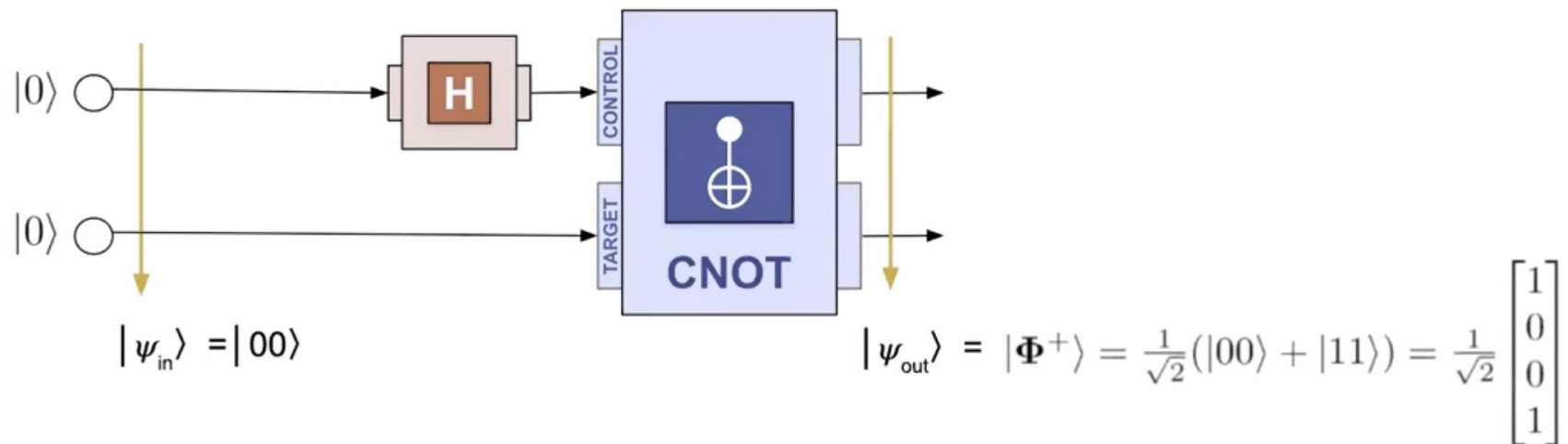
Applications of quantum computers

Quantum Teleportation

Step 1: Create an entangled pair of qubits

Entangle two qubits using **same entangle without phase**

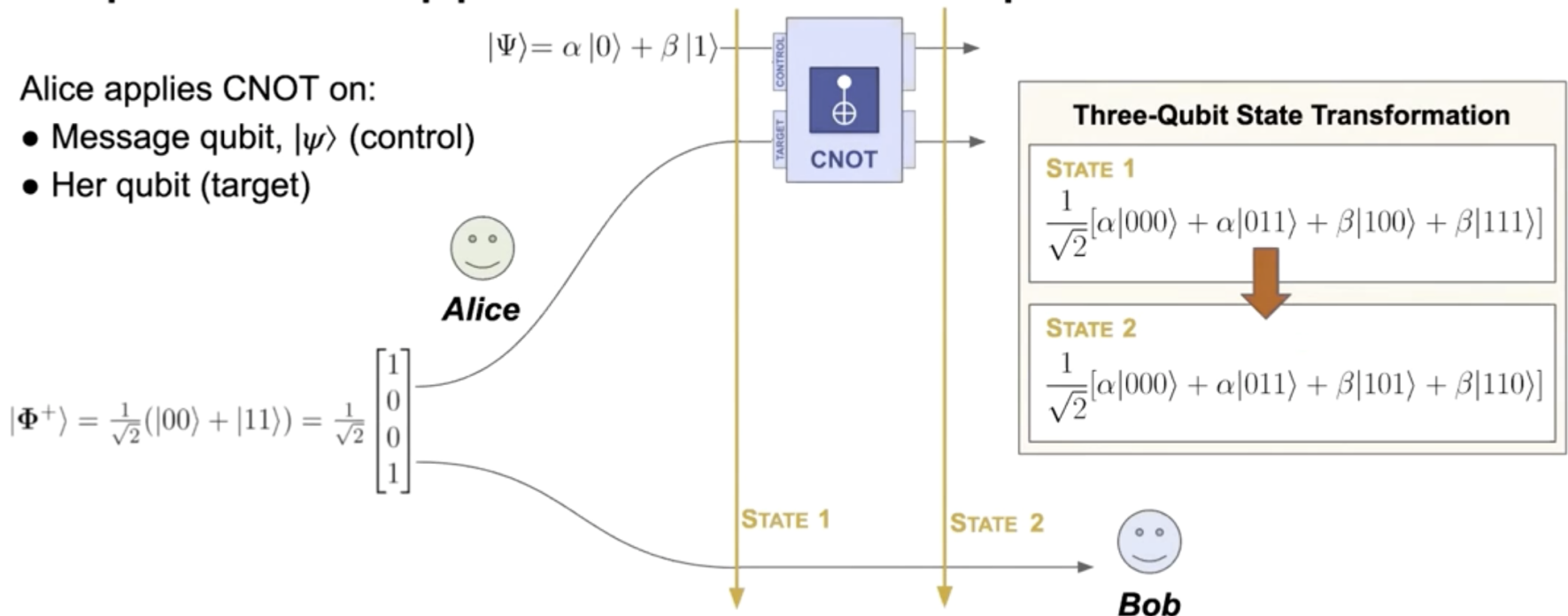
- Alice and Bob can entangle the qubits, or they can receive them from a third party!



Applications of quantum computers

Quantum Teleportation

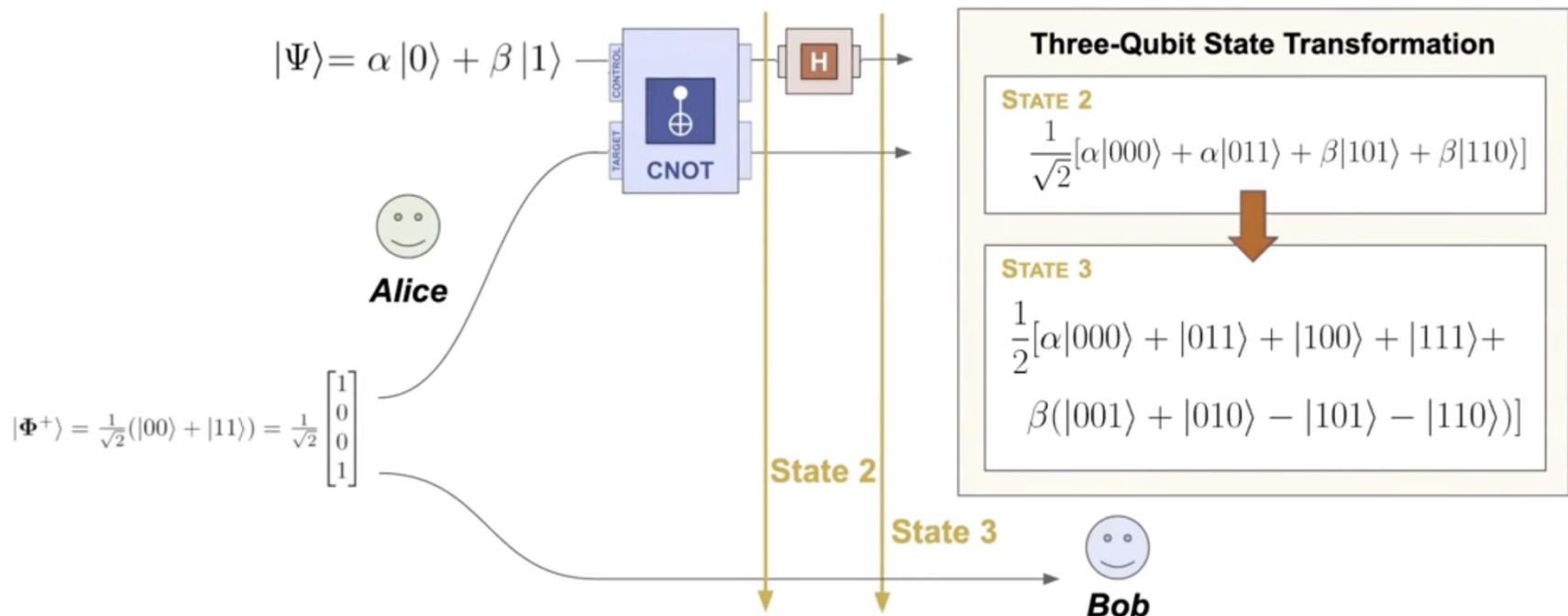
Step 3: Alice applies CNOT to her qubits



Applications of quantum computers

Quantum Teleportation

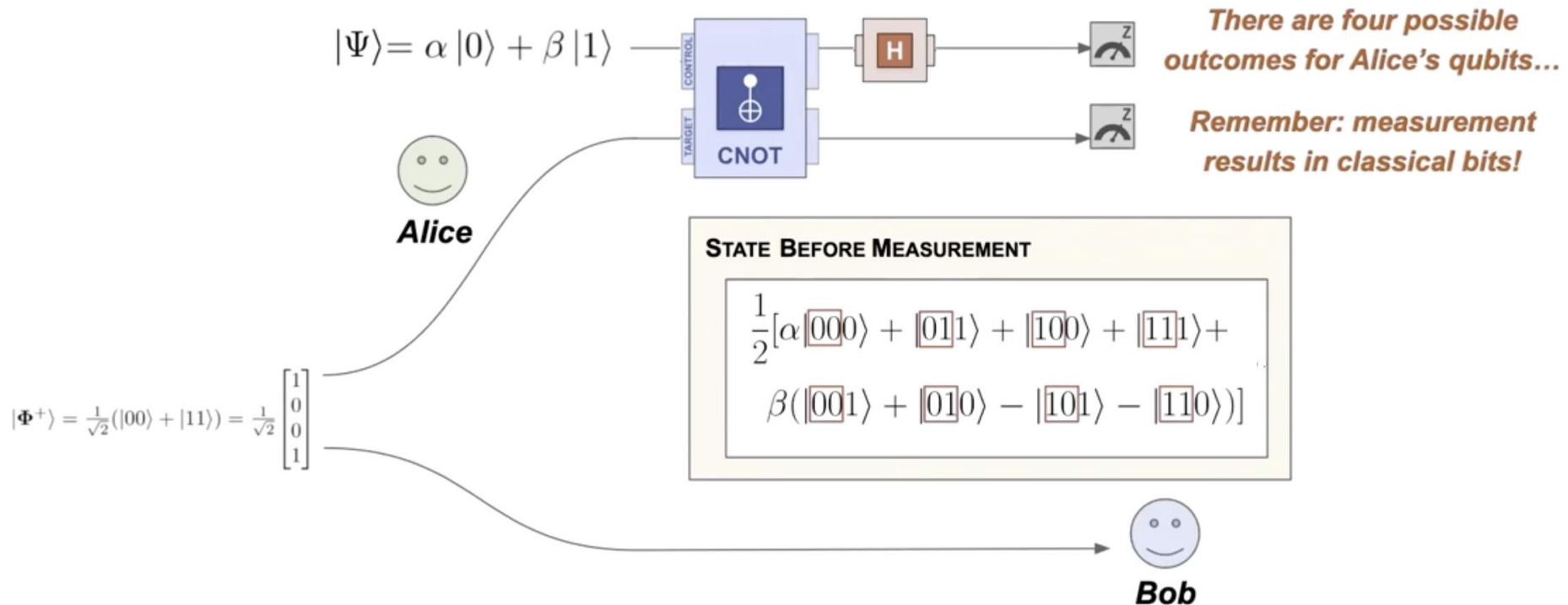
Step 4: Alice applies an H gate on $|\psi\rangle$



Applications of quantum computers

Quantum Teleportation

Step 5: Alice measures both of her qubits



Applications of quantum computers

Quantum Teleportation

Step 6: Process results of measurement

We deduce information about Bob's state by using partial measurements!

State before Measurement:

$$\frac{1}{2}[\alpha|000\rangle + |011\rangle + |100\rangle + |111\rangle + \beta(|001\rangle + |010\rangle - |101\rangle - |110\rangle)]$$

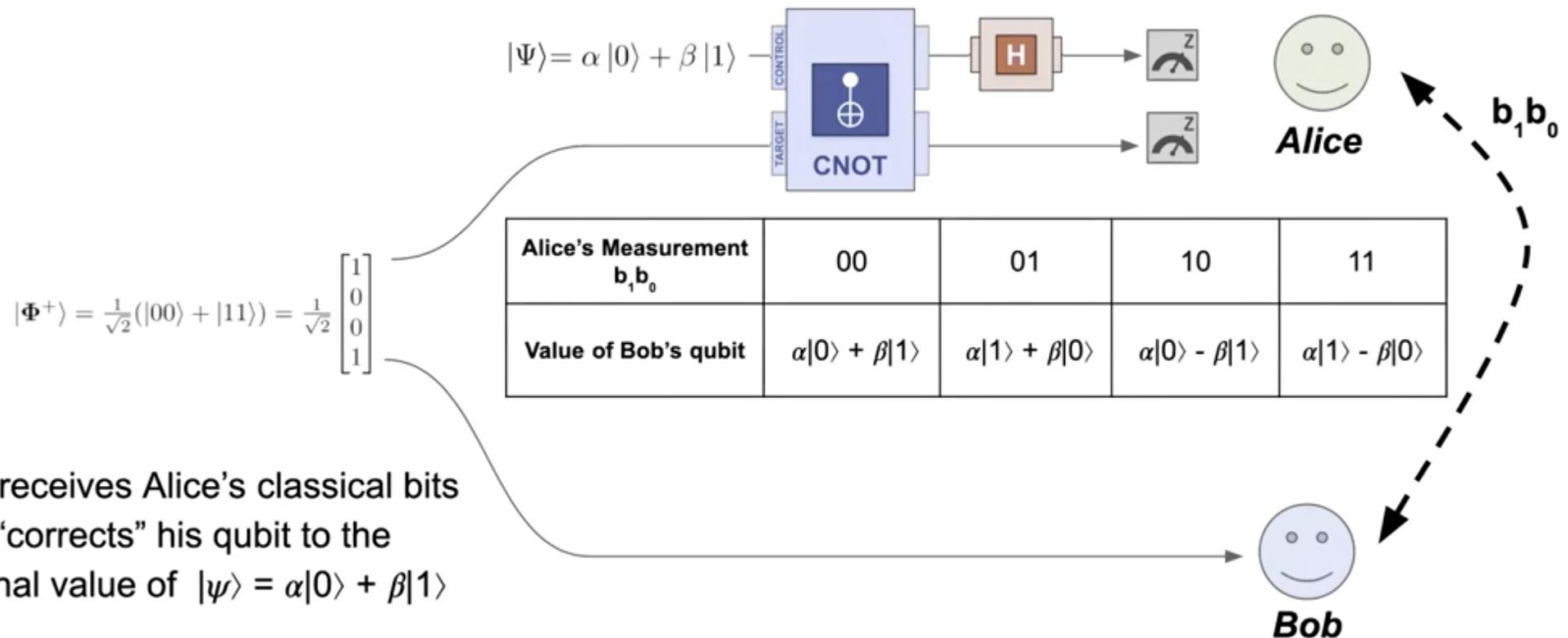
For example, if Alice sees 00, we narrow down the state of the entire system to the possibilities that satisfy this observation: $|000\rangle$ and $|001\rangle$.

Alice's Measurement	00	01	10	11
Value of Bob's qubit	$\alpha 0\rangle + \beta 1\rangle$	$\alpha 1\rangle + \beta 0\rangle$	$\alpha 0\rangle - \beta 1\rangle$	$\alpha 1\rangle - \beta 0\rangle$

Applications of quantum computers

Quantum Teleportation

Step 7: Alice transmits her two classical bits to Bob



How does Bob correct the state of $|\psi\rangle$?

Applications of quantum computers

Quantum Teleportation

Step 8: Bob recovers $|\psi\rangle$

To recover $|\psi\rangle$:

- if b_1 is 1, then apply a Z gate
- if b_0 is 1, then apply a NOT (X) gate

Alice's Measurement $b_1 b_0$	00	01	10	11
Value of Bob's qubit	$\alpha 0\rangle + \beta 1\rangle$	$\alpha 1\rangle + \beta 0\rangle$	$\alpha 0\rangle - \beta 1\rangle$	$\alpha 1\rangle - \beta 0\rangle$

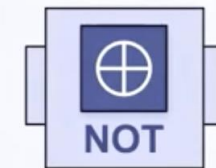
*No Gates
Needed!*

*Apply
NOT Gate*

*Apply
Z Gate*

*Apply
NOT and Z Gate*

Correction Gates:



$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

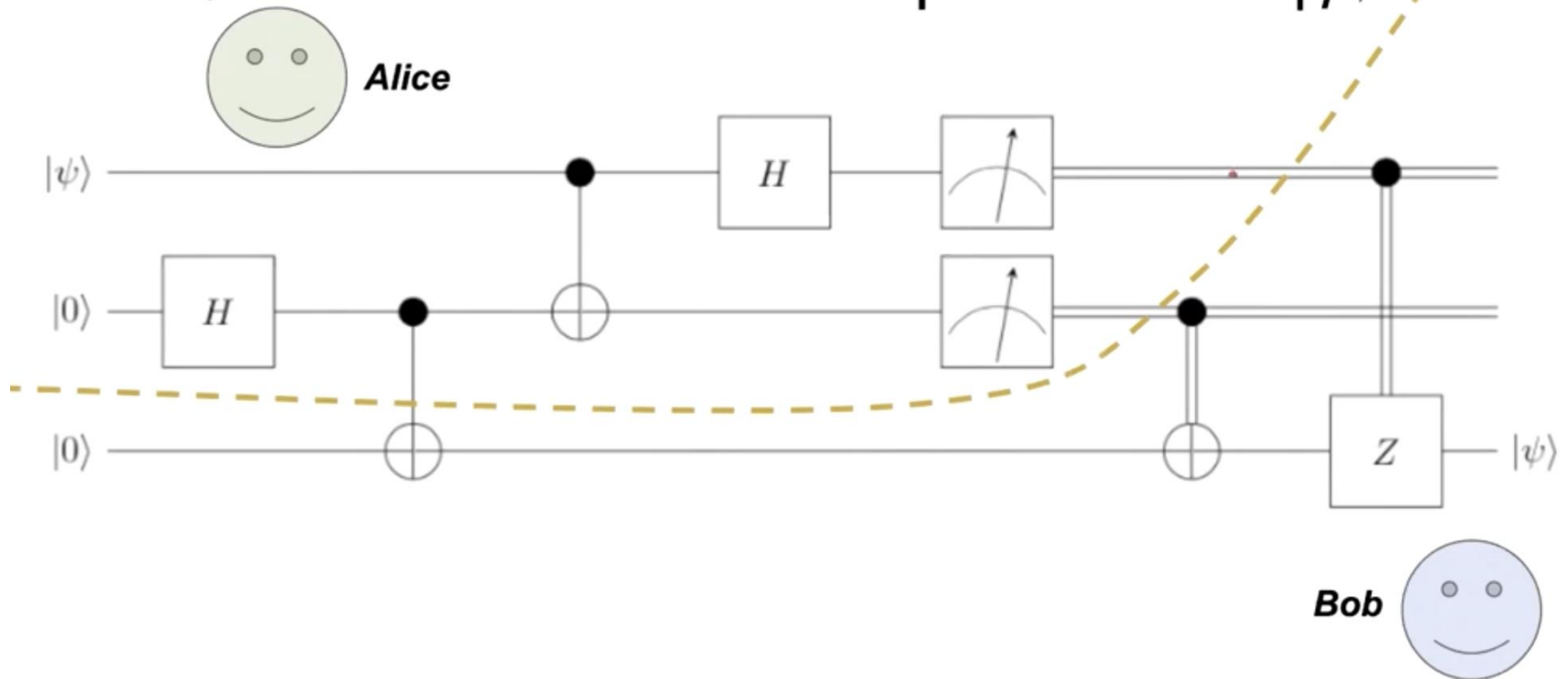


$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Applications of quantum computers

Quantum Teleportation

Full Quantum Circuit for Teleportation of $|\psi\rangle$

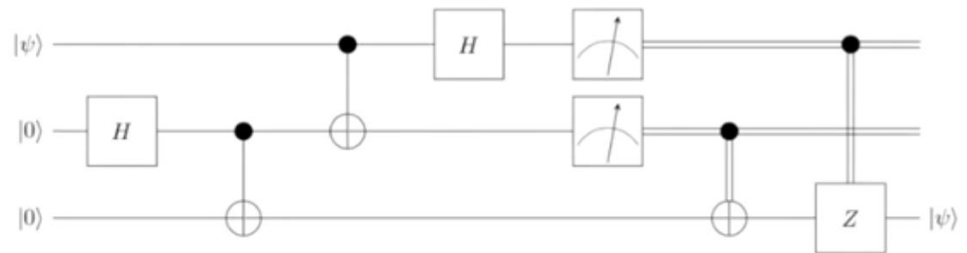


Applications of quantum computers

Quantum Teleportation

Notes about Quantum Teleportation

- Alice never knows the state of $|\psi\rangle$
- Communication is not faster than light because of the need for a classical communication channel
- Alice and Bob destroy their entangled qubits during the process of teleportation... more arbitrary qubit transfer requires more distributed entanglement!
- Quantum teleportation has been experimentally demonstrated many times
 - Good example of where quantum computing has “advantage” over classical methods
 - [In 2017, qubits were teleported from earth to space!](#)



Applications of quantum computers

Quantum Teleportation

- Discovered in 1993
- Experimentally realised in 1997
- The latest reported record distance for quantum teleportation is 1,400 km (870 miles) using the Micius satellite for space-based quantum teleportation



zmescience.com

Micius Satellite

Applications of quantum computers

Quantum Teleportation

Why is Teleportation Beneficial?

- 1) The actual transmission is faster using classical lines than quantum lines
 - “Moving” qubit state over distances needed for communication is very slow and error prone.
- 2) Moving **known** qubit states is more reliable than **unknown** states
 - Mobile qubits are in the same entangled state. Can pre-share many entangled qubits in advance for later teleportation procedures.
- 3) Option to use parallelization during teleportation
 - “Slow” qubit is transmitted at the same time as its pair does quantum calculations. The resulting delay is only the classical line transmission and gates to reproduce teleported qubit at destination.

Applications of quantum computers

Superdense coding: Alice's transmission

Superdense coding was discovered in 1992, and experimentally realised in 1996, it goes as follows:

Alice and Bob share an entangled pair, Alice wants to send two bits, i.e., one of 00, 01, 10 or 11. To do so, she applies a single-qubit unitary to her qubit:

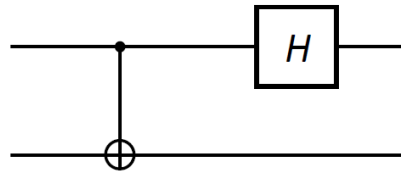
Initial state	Alice's bitstring	Operation	Final state
$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	00	I	$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$
$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	01	X	$\frac{1}{\sqrt{2}}(10\rangle + 01\rangle)$
$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	10	Z	$\frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$
$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	11	XZ	$\frac{1}{\sqrt{2}}(10\rangle - 01\rangle)$

Alice then sends her qubit to Bob.

Applications of quantum computers

Superdense coding: Bob's correction

Bob then receives Alice's qubit, so now has both qubits, and applies the following circuit to the two:



Which yields (ignoring a global phase factor of -1 after the CNOT in the last line):

Initial state	After CNOT	After H
$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	$\frac{1}{\sqrt{2}}(00\rangle + 10\rangle) = \frac{1}{\sqrt{2}}(0\rangle + 1\rangle) 0\rangle$	$ 00\rangle$
$\frac{1}{\sqrt{2}}(10\rangle + 01\rangle)$	$\frac{1}{\sqrt{2}}(11\rangle + 01\rangle) = \frac{1}{\sqrt{2}}(0\rangle + 1\rangle) 1\rangle$	$ 01\rangle$
$\frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$	$\frac{1}{\sqrt{2}}(00\rangle - 10\rangle) = \frac{1}{\sqrt{2}}(0\rangle - 1\rangle) 0\rangle$	$ 10\rangle$
$\frac{1}{\sqrt{2}}(10\rangle - 01\rangle)$	$\frac{1}{\sqrt{2}}(11\rangle - 01\rangle) = \frac{1}{\sqrt{2}}(0\rangle - 1\rangle) 1\rangle$	$ 11\rangle$

The final step is for Bob to perform a computational basis measurements on the two qubits, which will give him Alice's bitstring.

Applications of quantum computers

Quantum key distribution

- Quantum key distribution (QKD) was discovered in 1984, and the original protocol (which we will study) is known as BB84 after its discoverers, Charles Bennett and Gilles Brassard .
- It later turned out that QKD had previously been discovered, but not made public, by researchers at GCHQ.
- BB84 does not require entanglement (although some subsequent protocols do).



news.sky.com

GCHQ

- The record bit rate (of exchange of secure keys) is 1 Mbit/s, in a collaboration between this university and Toshiba.

Applications of quantum computers

Quantum key distribution

The one-time pad

These days, we are used to public-key cryptography, such as RSA which relies on the one-way nature of some mathematical function (i.e., factoring numbers is hard – or is it?!) to *computationally* guarantee security. A stronger requirement is to absolutely (*information theoretically*) guarantee security. Of which the simplest example is a one-time pad:

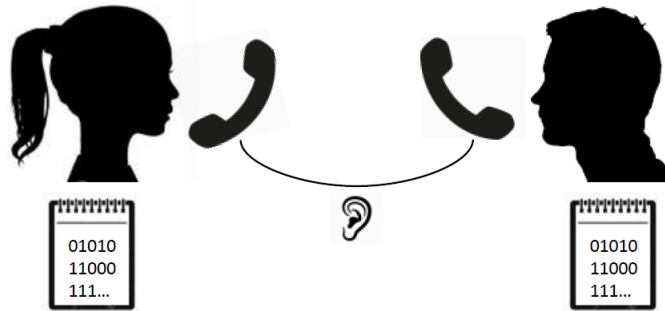
- At some date in the future Alice will send Bob an n bit message.
- Before that Alice and Bob meet-up and share a “one-time pad” (or key) – a list of n random bits r .
- When the time comes to send the message m , Alice encodes the message by using her copy of r to send $m \oplus r$.
- Bob receives the message and decodes it by using his copy of r :
$$(m \oplus r) \oplus r = m$$

Alice and Bob then discard r .

Applications of quantum computers

Quantum key distribution

Resources required to use a one-time pad



Lets take a more detailed look at the practicalities of using a one-time pad:

1. Alice and Bob must previously meet in person (or communicate at a distance via an absolutely secure channel).
2. Alice sends an encoded message, $m \oplus r$ (that is, $m + r \bmod 2$), to Bob via a channel, which in general could be tapped...

...but without access to r , all that an eavesdropper (Eve) would get is a random string of bits.

So the problem here is item 1, that Alice and Bob must meet in advance (or that they must have an absolutely secure channel – in which case they may as well use that for the message transmission).

Applications of quantum computers

Quantum key distribution

A one-time pad from quantum key distribution

QKD can be used to generate a one-time pad without Alice and Bob meeting, the resources required to achieve this are:

- An authenticated public classical channel. By “authenticated” we mean that if a transmission purports to be from Bob, then Alice can be absolutely sure it was indeed sent by Bob (and vice versa).
- A quantum channel, which could possibly be eavesdropped.

Additionally,

- **Alice** has a private source of random classical bits.
- **Alice** can produce qubits in states $|0\rangle$, $|1\rangle = X|0\rangle$, $|+\rangle = H|0\rangle$ and $|-\rangle = H|1\rangle$.
- **Bob** can measure qubits in either the computational ($|0\rangle, |1\rangle$) basis, or the $|+\rangle, |-\rangle$ basis.

Applications of quantum computers

Quantum key distribution

The BB84 protocol

1. Alice has a bitstring, and for each bit she either encodes $\{0, 1\}$ as $\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$ (chosen at random with equal probability). Alice then sends the qubit to Bob.
2. Bob receives the qubit and either measures in the $\{|0\rangle, |1\rangle\}$ basis or the $\{|+\rangle, |-\rangle\}$ basis.
3. Bob announces over a public channel in which basis he measured the qubit.
4. Alice replies over the public channel whether that was the basis in which the qubit was prepared.
5. If the same basis was indeed used for the preparation and the measurement then Bob's measurement outcome will equal Alice's bit, and they both append this bit to each of their copies of the key, otherwise they discard.

On average, Alice and Bob will discard half of their bits. In the following worked example we will see that this does indeed yield a shared key, and furthermore we will see that **it is private in the sense that any attempt by a third-party to discover the key will lead to a detectable change.**

Applications of quantum computers

Quantum key distribution

BB84: worked example

A bit	A basis	Qubit	B basis	B bit
0	$ 0\rangle, 1\rangle$	$ 0\rangle$	$ 0\rangle, 1\rangle$	0
1	$ +\rangle, -\rangle$	$ -\rangle$	$ +\rangle, -\rangle$	1
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$
0	$ 0\rangle, 1\rangle$	$ 0\rangle$	$ 0\rangle, 1\rangle$	0
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ 0\rangle, 1\rangle$	1
0	$ +\rangle, -\rangle$	$ +\rangle$	$ +\rangle, -\rangle$	0
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$
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Applications of quantum computers

Quantum key distribution

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0	$ +\rangle, -\rangle$	$ +\rangle$	$ +\rangle, -\rangle$	0
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$
1	$ +\rangle, -\rangle$	$ -\rangle$	$ 0\rangle, 1\rangle$	$\{0, 1\}$

The shared key (one-time pad) is the bits for which A and B agree on the measurement basis.

Applications of quantum computers

Quantum key distribution

BB84 attack : intercept, measure and retransmit

Which of the bits Alice and Bob transmitted / measured in the same basis is a matter of public record. However, it is also possible that an eavesdropper could “tap” the quantum channel to try to discover the key. The first option is for Eve to **intercept, measure and retransmit**.

However, owing to the fact that the protocol specifies that it is Bob who announces his measurement basis and Alice who then replies, it is necessary that Eve forwards on to Bob the intercepted qubit before it is made public which basis Alice transmitted in. So it follows that Eve would have to make a random decision about a basis to measure in. Moreover, the agreement about which qubits would be used in the key remains between Alice and Bob alone.

Applications of quantum computers

Quantum key distribution

BB84: worked example with eavesdropping

Consider the same example as before, but now with an eavesdropper between Alice and Bob.

A bit	A basis	Qubit	E basis	E bit	Qubit	B basis	B bit
0	$ 0\rangle, 1\rangle$	$ 0\rangle$	$ 0\rangle, 1\rangle$	0	$ 0\rangle$	$ 0\rangle, 1\rangle$	0
1	$ +\rangle, -\rangle$	$ -\rangle$	$ 0\rangle, 1\rangle$	$\{0, 1\}$	$\{ 0\rangle, 1\rangle\}$	$ +\rangle, -\rangle$	$\{0, 1\}$
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$	$\{ +\rangle, -\rangle\}$	$ +\rangle, -\rangle$	$\{0, 1\}$
0	$ 0\rangle, 1\rangle$	$ 0\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$	$\{ +\rangle, -\rangle\}$	$ 0\rangle, 1\rangle$	$\{0, 1\}$
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ 0\rangle, 1\rangle$	1	$ 1\rangle$	$ 0\rangle, 1\rangle$	1
0	$ +\rangle, -\rangle$	$ +\rangle$	$ +\rangle, -\rangle$	0	$ +\rangle$	$ +\rangle, -\rangle$	0
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ 0\rangle, 1\rangle$	1	$ 1\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$
1	$ +\rangle, -\rangle$	$ -\rangle$	$ 0\rangle, 1\rangle$	$\{0, 1\}$	$\{ 0\rangle, 1\rangle\}$	$ 0\rangle, 1\rangle$	$\{0, 1\}$

Applications of quantum computers

Quantum key distribution

BB84: worked example with eavesdropping

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1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ 0\rangle, 1\rangle$	1	$ 1\rangle$	$ 0\rangle, 1\rangle$	1
0	$ +\rangle, -\rangle$	$ +\rangle$	$ +\rangle, -\rangle$	0	$ +\rangle$	$ +\rangle, -\rangle$	0
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ 0\rangle, 1\rangle$	1	$ 1\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$
1	$ +\rangle, -\rangle$	$ -\rangle$	$ 0\rangle, 1\rangle$	$\{0, 1\}$	$\{ 0\rangle, 1\rangle\}$	$ 0\rangle, 1\rangle$	$\{0, 1\}$

As before, Alice and Bob expect to use the bits for which they agree on the measurement basis as the shared key.

Applications of quantum computers

Quantum key distribution

BB84: worked example with eavesdropping

Consider the same example as before, but now with an eavesdropper between Alice and Bob.

A bit	A basis	Qubit	E basis	E bit	Qubit	B basis	B bit
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1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$	$\{ +\rangle, -\rangle\}$	$ +\rangle, -\rangle$	$\{0, 1\}$
0	$ 0\rangle, 1\rangle$	$ 0\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$	$\{ +\rangle, -\rangle\}$	$ 0\rangle, 1\rangle$	$\{0, 1\}$
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ 0\rangle, 1\rangle$	1	$ 1\rangle$	$ 0\rangle, 1\rangle$	1
0	$ +\rangle, -\rangle$	$ +\rangle$	$ +\rangle, -\rangle$	0	$ +\rangle$	$ +\rangle, -\rangle$	0
1	$ 0\rangle, 1\rangle$	$ 1\rangle$	$ 0\rangle, 1\rangle$	1	$ 1\rangle$	$ +\rangle, -\rangle$	$\{0, 1\}$
1	$ +\rangle, -\rangle$	$ -\rangle$	$ 0\rangle, 1\rangle$	$\{0, 1\}$	$\{ 0\rangle, 1\rangle\}$	$ 0\rangle, 1\rangle$	$\{0, 1\}$

As before, Alice and Bob expect to use the bits for which they agree on the measurement basis as the shared key.

However, the presence of the eavesdropper has now randomised some of Bob's measurement outcomes. It follows that a comparison between Alice and Bob's measurement outcomes for *some* of the bits where they have the same measurement basis suffices (statistically) to detect the eavesdropper, and thus indicates that the rest of the bits have been compromised, and cannot be used as the key.

Applications of quantum computers

Quantum key distribution

Eavesdropping – other factors

As seen in the worked example, eavesdropping disturbs the shared key – thus whilst Alice and Bob can rest assured that Eve hasn't discovered their key, they do need to set aside a subset of the bits to compare on the public channel, to check whether their key has been disturbed by eavesdropping.

Eve may try to avoid the problem by using a more sophisticated **intercept, copy, retransmit** attack, where she would keep a copy of the qubit, and only measure it once Alice and Bob had shared on the public channel the bases they agreed on – but this would **violate the no-cloning principle**.

Another attack could involve Eve storing (not copying) the intercepted qubit, and forwarding on a pre-prepared random qubit to Bob. Bob and Alice would then publicly share their measurement bases, and so Eve could indeed use this information to perform a measurement on the intercepted qubit and thus discover the key. However, as Bob would simply have a load of random bits, the public comparison of measurement outcomes that Alice and Bob conduct on some of the bits for which they agree the measurement basis would reveal the presence of the eavesdropper.

Applications of quantum computers

Case Study 2: Shor Algorithm of Factoring

- Theorem of Number Theory
 - If $a \not\equiv \pm b \pmod{N}$ but $a^2 \equiv b^2 \pmod{N}$, then $\gcd(a+b, N)$ is a factor of N .
- To Factor N on a quantum computer:
 - Select x coprime to N
 - Use QFT on quantum computer to find the period of $f(s) = x^s \pmod{N}$
 - Use order of x to compute possible factors of N .
 - Check if they work; If not, rerun.
- Best classical algorithm takes time $O(\exp(n^{1/3}))$
But Shor's quantum algorithm takes time $O(n^3 \log n)$



Peter Shor 1994

Applications of quantum computers

Quantum Fourier Transform (QFT)

$$U_{QFT} = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \sum_{x=0}^{N-1} \omega_N^{xy} |y\rangle \langle x|$$

$$U_{QFT}^\dagger = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \sum_{x=0}^{N-1} \omega_N^{-xy} |x\rangle \langle y|$$

$$|\psi(x_0)\rangle = \frac{1}{\sqrt{2^n/l}} \sum_{j=0}^{2^n/l-1} |x_0 + jl\rangle$$

$$\begin{aligned} & \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \left\{ \begin{array}{c} \text{---} \\ \vdots \\ \text{---} \end{array} \right\} \left[\begin{array}{c} \text{---} \\ \vdots \\ \text{---} \end{array} \right] U_f \left[\begin{array}{c} \text{---} \\ \vdots \\ \text{---} \end{array} \right] |f(x_0)\rangle \\ & \quad |0\rangle \left\{ \begin{array}{c} \text{---} \\ \vdots \\ \text{---} \end{array} \right\} \\ & \quad f(s) = x^s \bmod N \end{aligned}$$

$$U_{QFT} \left[\begin{array}{c} \text{---} \\ \vdots \\ \text{---} \end{array} \right] = \frac{1}{\sqrt{2^{2^n/l}}} \sum_{p=0}^{2^n-1} \sum_{j=0}^{2^n/l-1} \exp\left(\frac{2\pi i p(x_0 + jl)}{2^n}\right) |p\rangle$$

$$= \frac{1}{\sqrt{l}} \sum_{p=0}^{2^n-1} \sum_{|pl=0 \bmod 2^n} e^{\frac{2\pi i p x_0}{2^n}} |p\rangle$$

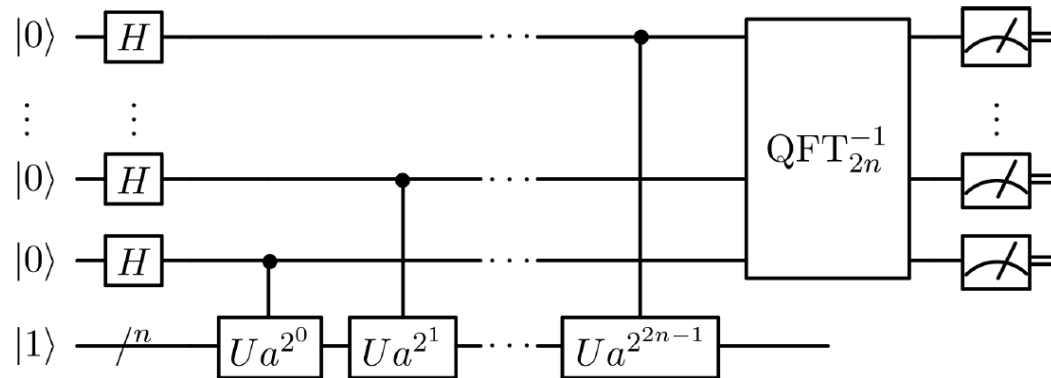
Applications of quantum computers

Case Study 2: Shor Algorithm of Factoring

18819881292060796383869723946165043	=	3980750864240649373971	×	4727721461074353025362
98071635633794173827007633564229888				
59715234665485319060606504743045317				
38801130339671619969232120573403187				
9550656996221305168759307650257059				

2550055038649119906436
2342526708406385189575
946388957261768583317

2307197304822463291469
5302097116459852171130
520711256363590397527



Applications of quantum computers

Case Study 3: Variational Quantum Eigensolver

- VQE is a quantum/classical hybrid algorithm that can be used to find eigenvalues of a matrix H .
- When VQE is used in quantum simulations, H is typically the Hamiltonian of an electronic system.
- It is run inside of a classical optimization loop.
- The quantum part has two fundamental steps:
 - Prepare the quantum states $|\psi\rangle$, often called the ansatz;
 - Measure the expectation value $\langle \psi | H | \psi \rangle$
- The classical optimization loop does two tasks:
 - Use a classical non-linear optimizer to minimize the expectation value by varying ansatz parameters
 - Iterate until convergence is reached.

Applications of quantum computers

Post-Quantum World

There is a good chance that quantum computers will be able to crack RSA-2048 within five-ten years (need about 8,000 qubits in a universal computer to do this).

Some encrypted data has a shelf-life of more than ten years. It may take ten years to cut over to a new encryption scheme, so companies and governments are scrambling to figure out what to do.

Most encryption used today is not safe in a quantum world.

If someone has been recording a https session, say, they may not be able to decrypt it now, but a few years from now, who knows.

There are encryption algorithms that are safe:

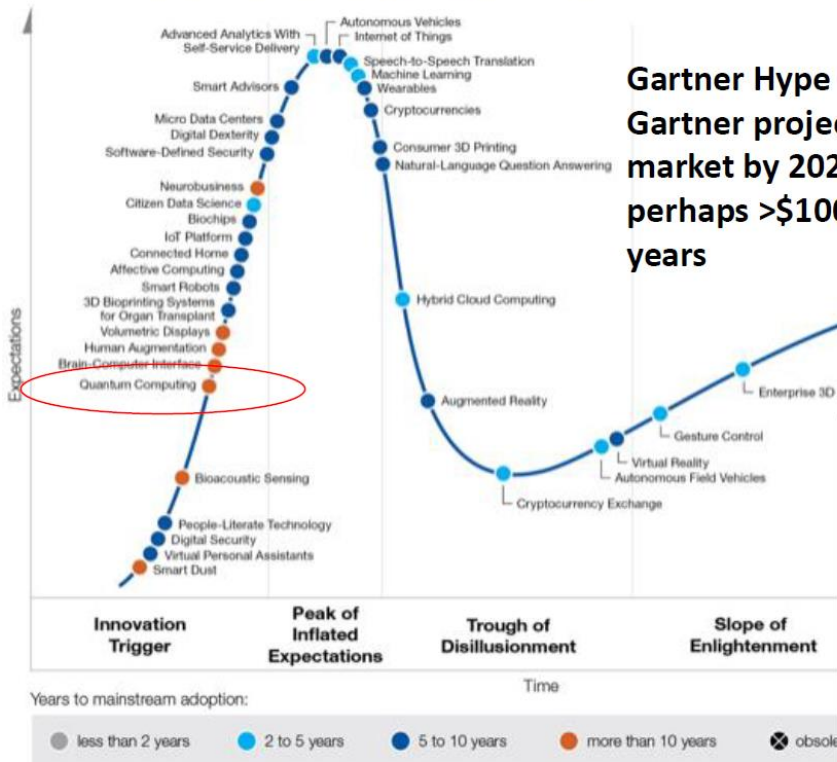
Secure communications will move to symmetric encryption or hash algorithms, or perhaps lattice algorithms.

QKD (Quantum Key Distribution) will likely be more popular for the most sensitive communications

Some governments and banks are already starting to use “Quantum Safe” algorithms, and this is expected to increase over the next decade.

Applications of quantum computers

Emerging Technology Hype Cycle

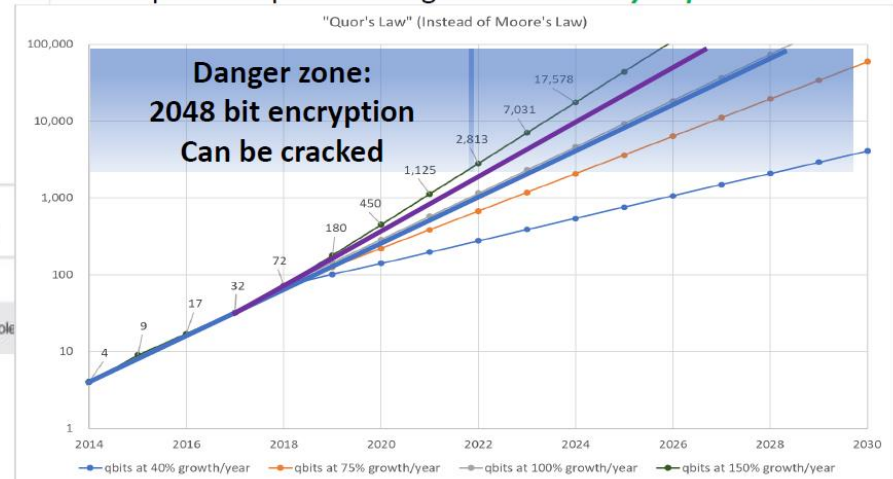


Gartner Hype Cycle.
Gartner projects a \$5B market by 2020 worth perhaps >\$100B in ten years

Future of Quantum Computers

*Qubit count of general-purpose quantum computers will double roughly every one to two years, or roughly every **18 months** on average.*

Neven's Law, named after Hartmut Neven, the director of Google's Quantum Artificial Intelligence Lab, stated that quantum processors grow at a **doubly-exponential rate**.



Applications of quantum computers

Problems with Quantum Computers

- Decoherence
 - Quantum system is extremely sensitive to external environment, so it should be safely isolated
 - It is hard to achieve the decoherence time that is more than the algorithm running time
- Error correction (requires more qubits!)
- Physical implementation of computations
- New quantum algorithms to solve more problems
- Entangled states for data transfer
- Running time increases exponentially with matrix size